Sahil Khan

AI / ML ENGINEER

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Portfolio LinkedIn GitHub

Summary

AI/ML Engineer with 1+ year of hands-on experience developing real-world AI/ML and Generative AI projects, including Agentic AI, RAG systems, and end-to-end machine learning pipelines. Proficient in Python, LangChain, LangGraph, NLP, and LLM integration, with practical experience in deploying scalable applications using FastAPI, Docker, AWS, MLflow, and CI/CD.

Skills

Programming: Python, SQL, C++

AI/ML: Scikit-learn, PyTorch, Feature Engineering, Model Evaluation, Cross-Validation, NLP

Generative AI: RAG, Prompt Engineering, Embeddings, Vector Search (FAISS, Chroma), LangChain, LangGraph, LLM Evals

Agentic AI: LLM Gateway, LLM guardrails, Agent Evals, Tool Calling, LangSmith, Human-in-the-Loop, MCP

Backend & MLOps: FastAPI, Flask, REST APIs, Docker, AWS, DVC, MLflow, Git, GitHub Actions, CI/CD

Experience

AI Engineer Intern May 2025 – July 2025

Zetheta Algorithms Pvt. Ltd. Python, NLP, Scikit-learn, Pandas

- Developed an AI-powered intern–employer matchmaking platform to convert unstructured resumes into standardized candidate profiles, supporting internship matching for 1,000+ students.
- Built a resume-parsing pipeline using Named Entity Recognition (NER) and text preprocessing to extract skills, education, and experience from PDF and text files.
- Implemented candidate-job matching using TF-IDF and cosine similarity to measure alignment between candidate profiles and job requirements.
- Designed a 0–100 job-readiness scoring engine using weighted skill match, education relevance, and project experience to assess candidate suitability.

Projects

YouTube Comment Insights Engine | GitHub *Python, TF-IDF, LightGBM, Optuna, SMOTE, DVC, MLflow, Docker, AWS*

- Developed a three-class sentiment classification system (Positive, Neutral, Negative), achieving **87% accuracy** using LightGBM with TF-IDF unigram, bigram, and trigram features (up to 10,000 features).
- Engineered an NLP preprocessing pipeline with text normalization, stop-word removal, and lemmatization; experimented with Bag-of-Words vs. TF-IDF vectorization, feature selection, and SMOTE-based class balancing.
- Benchmarked multiple classifiers, including Logistic Regression, SVM, KNN, Naive Bayes, Random Forest, XGBoost, and LightGBM; leveraged Optuna for hyperparameter optimization and MLflow for experiment tracking and evaluation.
- Built a reproducible DVC pipeline for data ingestion, preprocessing, model training, evaluation, and registration; developed Flask REST APIs for sentiment predictions and visual analytics, with automated CI/CD testing and Docker image publishing to AWS ECR using GitHub Actions.

Real Estate Intelligence | GitHub *Python, Pandas, Scikit-learn, BeautifulSoup, Streamlit, Plotly, NumPy*

- Built an end-to-end Gurgaon real estate intelligence platform by scraping property listings from 99acres, covering flats, independent houses, residential land, and apartment projects.
- Engineered data preprocessing and feature engineering pipelines involving missing-value imputation, outlier treatment, feature selection, and extraction of property attributes including usable area, possession age, furnishing, luxury, and floor categories.
- Compared regression approaches and developed a Scikit-learn preprocessing and prediction pipeline using log-transformed property prices, with inverse transformation via `expm1` for price estimation.
- Developed a multi-page Streamlit application with Plotly-based geospatial market analytics, sector-wise price-per-square-foot visualizations, and an apartment recommendation engine ranking the **top 7 similar properties** using weighted cosine similarity across three feature-based similarity matrices.

Education

Indian Institute of Technology, Dhanbad July 2024 – May 2026

M.Sc. in Mathematics and Computing Dhanbad, India

Relevant Coursework: Data Structures and Algorithms, Data Mining, Big Data, DBMS

Achievements

- Qualified IIT JAM 2024 in Mathematics with All India Rank 617.
- Qualified CUET PG 2024 in Mathematics.